

Modeling and **model-assisted prospective** for forests and forest sectors

3.3

Course unit 3.3 — opening session, Friday 4 September 2026.

COURSE LEADERS

A. Lobianco · V. Mathieu

CONTACT HOURS

54 h

BLOCKS

Sept · Oct · Dec · Jan

ASSESSED DELIVERABLE

15 January

THE QUESTION THIS COURSE OPENS WITH

How many models will you meet before March?

Around a dozen — across almost all course units, several external speakers, four half-days of expert-led sessions. Documenting them, together, is **40 % of this course unit**.

Where this course sits

A transversal unit, spread across the whole first semester. It is not a block you take and close.

1

Across the semester

Four blocks from September to January, inside UE 3 — Policies and governance of forest resources. Lectures, hands-on tutorials, external experts, group work.

TRANSVERSAL

2

It feeds the three engineering projects

Framing a modelling problem, choosing a suitable model, implementing it reproducibly, handling uncertainty, communicating projections — all of it lands directly in your project deliverables.

DIRECTLY REUSABLE

3

And it underpins every other module

Modelling, scientific programming and the good practice around them come up in most other units. These materials stay your reference — come back to them whenever another course puts a model or a script in front of you.

ALL YEAR LONG

What you should be able to do in March

01 Design and implement quantitative models

On complex forest questions — growth, forest sector, carbon...

02 Work with reproducible scientific programming

Python, R, Julia, git, environments, workflows... — to develop and run models.

03 Analyse and compare the main families of models

Partial equilibrium, agent-based, process-based, empirical / ML, spatially explicit... — making assumptions, strengths and limits explicit.

04 Build scenarios and quantify uncertainty

Prospective scenarios, sensitivity analysis, validation.

05 Interpret and communicate model-based projections

To non-expert audiences, in support of decision-making and forest policy.

Four blocks, September to January

WHEN	CONTENT	FORMAT	WHO
BLOCK 1 — FUNDAMENTALS AND PROGRAMMING BASICS · SEPTEMBER			
4 Sept.	Fundamentals of modelling — research question, typology, scenarios, uncertainty, validation, coupling, reproducibility	Lecture 3 h	A. Lobianco
7 Sept.	Foundations of scientific programming	Lecture 3 h · Lab 3 h	V. Mathieu
BLOCK 2 — MODELLING TECHNIQUES AND ADVANCED PROGRAMMING · OCTOBER			
20–23 Oct.	Julia, environments, git / GitHub, reproducible workflows... — plus two half-days of project time on the cartography	Lecture 1 h 30 · Lab 10 h 30 · Project 6 h	A. Lobianco · V. Mathieu
BLOCK 3 — SPATIALLY EXPLICIT MODEL · DECEMBER			
16–18 Dec.	Spatially explicit forest resource modelling — EFISCEN Space, applied cases (storms, restoration), plus a seminar	Seminar 3 h · Lab 12 h	G.-J. Nabuurs (WUR)
BLOCK 4 — PORTFOLIO OF REPRESENTATIVE MODELS · OCTOBER → JANUARY			
4 half-days	Expert-led sessions on representative models (speakers being confirmed)	Seminars 4 × 3 h	External speakers
THROUGHOUT THE YEAR			
All year	Group work — cartography of the models covered across the FIRS programme	Personal work	V. Mathieu · A. Lobianco

54 contact hours in total, of which 23 hours of teaching. Language of instruction: English.

Ground rules

ATTENDANCE AND GROUP WORK

- A

All sessions are compulsory

Lectures, labs, external contributions. Any absence must be justified promptly. Active participation counts.
- B

You work in pairs, formed by the teaching team

Two students per group, put together so that the pairs stay balanced. We will come back to this on 7 September.
- C

The deliverable is everyone's

Tasks split fairly, but each of you answers for the whole thing. Tell us early if a group stops working.

DEADLINES AND INTEGRITY

- D

Deadlines hold

Flag a delay in advance. Otherwise a penalty applies (typically –10 % per day), and beyond a limit the work may be refused.
- E

Academic integrity

No plagiarism, no fabricated data, no unauthorised collusion. Cite every source — articles, data, code, materials — and be transparent about your use of AI tools.

How you are assessed

60%

Technical deliverables of the engineering projects — the best two out of three, with progression taken into account. GitHub repository or equivalent.

40%

Cartography of the models covered across the FIRS programme, with a 3-page synthesis note. Group work.

WHAT WE LOOK AT

- Technical rigour and relevance
- Quality and reproducibility of the code
- Demonstrated progression from one project to the next
- Coverage and structure of the cartography
- Critical and comparative analysis
- Clarity of the write-up

The cartography of models

What	A cartography of the models covered across the whole FIRS programme Not only the ones seen in this course unit.
Form	The cartography itself, plus a 3-page synthesis note (PDF)
Who	Pair work — two students per group, formed by the teaching team
When	15 January Built up throughout the year, not written in January.
Weight	40 % of the course unit

Ten expert sessions, or one reading grid

Without a shared reading grid, a dozen model presentations stay **a dozen anecdotes** — impressive on the day, unusable six months later.

The skill we are after: **choose a model that fits a question** — and be able to say what that model **cannot** do.

SCOPE

Which models go in

SEEN INSIDE THIS COURSE UNIT

EFISCEN Space

TiMBa

Margot

...

AND EVERY MODEL YOU MEET ELSEWHERE IN FIRS

Survey sampling and inference

Forest inventory and monitoring

Remote sensing

Data analytics

Climate change and forests

Wood markets

Your engineering projects

...

If a course puts a model in front of you, it belongs in the cartography. **Keep your eyes open in every unit**, not only in this one.

What such a grid could contain

Not a template to fill in. A first idea of the questions worth asking of any model.

Identity	The question it answers · the family of model it belongs to
Frame	Spatial scale · time step · time horizon
Flows	Inputs — data, drivers · outputs produced
Credibility	Where the parameters come from · how the model is validated · how uncertainty is handled
Use	Access and licence · strengths · limits · domain of validity

Right now these are **words, not yet concepts**. The next 2 h 45 are what make a relevant grid possible — and you will settle its structure yourselves, on 20 October.

How much detail? A worked example

A deliberately trivial model — standing volume of a forest area over 50 years.

Question	How does standing volume evolve over 50 years under constant management?
Family	Empirical, aggregated stock-and-flow: growth – harvest – mortality
Scale · step · horizon	One forest area · annual · 50 years
Inputs	Initial volume, growth rate, harvest rate, mortality rate
Outputs	Standing volume, year by year
Parameters	Fitted on national forest inventory plots, 1990–2020
Validation	Back-casting over 2000–2020
Uncertainty	None — a single deterministic trajectory, no interval
Access	Fifteen lines of R, fully readable
Strengths	Transparent, instant, easy to argue about
Limits · domain of validity	No species, no spatial detail, no climate effect. Only credible under a climate and a management regime close to those of the calibration period.

What separates a good cartography from a catalogue

1

Cover and structure

The same reading grid applied to every model, so that two entries can be compared without going back to the sources.

2

Be critical and comparative

Compare the models with each other — assumptions, strengths, limits, domains of use. Paraphrasing each model's own documentation is the failure mode. This is the criterion that carries the most weight.

3

Make it readable

The synthesis note is three pages: it argues a point, it does not summarise the cartography.

A **guidance note** will be circulated, with several existing model typologies to draw on. Read it **before October** — it needs time to settle.

How this gets built

Before Oct.	List the models, and start sorting them Keep a running list of every model shown in your course units — and start thinking about how you would sort them. Read the guidance note: existing typologies, to draw on rather than to copy.
20 Oct. pm	Pairs and structure Pairs formed, and the structure of your cartography settled together — then a first model documented collectively.
22 Oct. pm	Progress session With the models already met in the other course units.
All year	Document each model the day you meet it
15 Jan.	Cartography + 3-page synthesis note

An entry written up the same evening costs **twenty minutes**. The same entry, reconstructed in January, costs **a day**.

THIS AFTERNOON

The next 2 h 45 are not a preamble

They are what turns that list of headings into concepts you can actually use.

Family of model	Typology of models — and how to read a model’s anatomy
Scale · step · horizon	Framing a question: object, scale, resolution, horizon
Parameters · validation	Calibration, verification, validation — where you may trust a model
Uncertainty	Typology of uncertainty, sensitivity analysis, propagation
Horizon · use	Scenarios and prospective, coupling, and what a model does in a policy debate

Antonello Lobianco — **Fundamentals of modelling**. You now know what to listen for.

MONDAY 7 SEPTEMBER

Foundations of scientific programming

The tools you need for this deliverable, and for the three engineering projects. Install **R + RStudio**, **Python**, **VS Code**, **git** and create a **GitHub account** before then.